

$^{32}\text{S}(^3\text{He}, n\gamma)$  1974Gr19, 1974Be18, 1972Ca22

**1974Be18:** an 8-MeV  $^3\text{He}$  beam was provided from the ANU EN tandem accelerator and impinged on cooled sulphur targets.

Neutrons were detected using an NE213 liquid scintillator and  $\gamma$  rays were detected using a Ge(Li) detector and a NaI(Tl) detector. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $n\gamma$ -coin,  $n\gamma(\theta)$ . Deduced levels, J,  $\gamma$ -ray branching ratios; deduced lifetimes for 4 levels using DSAM.

**1974Gr19:** 8-12-MeV  $^3\text{He}$  beams were provided from the 7-MV Van de Graaff accelerator at the University of Freiburg and impinged on natural ZnS targets. Neutrons were detected using an annular NE213 liquid scintillator and  $\gamma$  rays were detected using a Ge(Li) detector and 5 NaI(Tl) detectors. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $n\gamma$ -coin, and  $n\gamma\gamma$ -coin. Deduced 10 levels, J,  $\gamma$ -ray branching ratios, and mixing ratios; deduced lifetimes for 5 levels using DSAM.

**1972Ca22:** an 8.5-MeV  $^3\text{He}$  beam was provided from the Oxford University tandem Van de Graaff generator and impinged on  $\text{Ag}_2\text{S}$  targets. Neutrons were detected using a NE102 liquid scintillator. The  $\gamma$  rays were detected using a Ge(Li) detector. Measured  $E_\gamma$ ,  $I_\gamma$ ,  $E_n$ ,  $I_n$ , and  $n\gamma(\theta)$ . Deduced lifetimes for 3 levels using DSAM.

 $^{34}\text{Ar}$  Levels

| E(level) <sup>†</sup> | J <sup>π</sup>    | T <sub>1/2</sub> | Comments                                                                                                                                                                                                                                                       |
|-----------------------|-------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0                   | 0 <sup>+</sup>    |                  |                                                                                                                                                                                                                                                                |
| 2091.0 4              | 2 <sup>+</sup>    | 139 fs 35        | J <sup>π</sup> : from the Adopted Levels. Spin=2 confirmed by $n\gamma(\theta)$ in 1974Gr19.<br>T <sub>1/2</sub> : from lifetime $\tau=200$ fs 50: weighted average of 330 fs 80 (1974Be18, DSAM), 200 fs 60 (1974Gr19, DSAM), and 150 fs 50 (1972Ca22, DSAM). |
| 3287.5 6              | 2 <sup>+</sup>    | 104 fs 42        | J <sup>π</sup> : from the Adopted Levels. Spin=2 confirmed by $n\gamma(\theta)$ in 1974Gr19.<br>T <sub>1/2</sub> : from lifetime $\tau=150$ fs 60: weighted average of 215 fs 70 (1974Be18, DSAM), 130 fs 60 (1974Gr19, DSAM), and 120 fs 65 (1972Ca22, DSAM). |
| 3873 3                | 0 <sup>+</sup>    | >416 fs          | J <sup>π</sup> : from the Adopted Levels. Spin=0 confirmed by $n\gamma(\theta)$ in 1974Gr19.<br>T <sub>1/2</sub> : from lifetime $\tau>600$ fs (1974Gr19, DSAM).                                                                                               |
| 4127.4 11             | (2 <sup>+</sup> ) | <243 fs          | J <sup>π</sup> : spin=2 from $n\gamma(\theta)$ , but did not exclude spin=3 or 4; $\pi=+$ from mirror level (1974Gr19).<br>T <sub>1/2</sub> : from lifetime $\tau<350$ fs (1974Be18, DSAM). Other: $\tau>400$ fs (1974Gr19, DSAM).                             |
| 4513.5 8              | 3                 | 201 fs 55        | J <sup>π</sup> : spin=3 from $n\gamma(\theta)$ (1974Gr19).<br>T <sub>1/2</sub> : from lifetime $\tau=290$ fs 80: weighted average of 390 fs 180 (1974Be18, DSAM), 260 fs 80 (1974Gr19, DSAM), and 300 fs 80 (1972Ca22, DSAM).                                  |
| 4631 <sup>‡</sup> 4   |                   |                  |                                                                                                                                                                                                                                                                |
| 4865 <sup>‡</sup> 4   |                   |                  |                                                                                                                                                                                                                                                                |
| 4967 <sup>‡</sup> 4   |                   |                  |                                                                                                                                                                                                                                                                |
| 5255 <sup>‡</sup> 4   |                   |                  |                                                                                                                                                                                                                                                                |
| 5542 <sup>‡</sup> 4   |                   |                  |                                                                                                                                                                                                                                                                |

<sup>†</sup> For E(level) below 4.6 MeV, from a least-squares fit to  $\gamma$ -ray energies.

<sup>‡</sup> From 1974Gr19 based on  $\gamma$  data.

 $\gamma(^{34}\text{Ar})$ 

| E <sub>i</sub> (level) | J <sub>i</sub> <sup>π</sup> | E <sub>γ</sub> <sup>†</sup> | I <sub>γ</sub> <sup>‡</sup> | E <sub>f</sub> | J <sub>f</sub> <sup>π</sup> | Mult. | δ       | Comments                                                                                                                                                                                                                                                                                           |
|------------------------|-----------------------------|-----------------------------|-----------------------------|----------------|-----------------------------|-------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2091.0                 | 2 <sup>+</sup>              | 2090.9 4                    | 100                         | 0.0            | 0 <sup>+</sup>              |       |         | E <sub>γ</sub> : weighted average of 2091.1 3 (1974Be18) and 2090.0 7 (1972Ca22).<br>A <sub>2</sub> =+0.48 9 (1974Be18).                                                                                                                                                                           |
| 3287.5                 | 2 <sup>+</sup>              | 1196.5 4                    | 89 5                        | 2091.0         | 2 <sup>+</sup>              | M1+E2 | +0.12 5 | E <sub>γ</sub> : weighted average of 1196.6 4 (1974Be18) and 1195.8 14 (1972Ca22).<br>I <sub>γ</sub> : weighted average of 92 5 (1974Be18), 90 7 (1972Ca22), and 85 5 (1974Gr19).<br>Mult., δ: D+Q from $n\gamma(\theta)$ (1974Gr19); M2 ruled out by RUL.<br>A <sub>2</sub> =+0.56 10 (1974Be18). |
|                        |                             | 3286 4                      | 11 5                        | 0.0            | 0 <sup>+</sup>              |       |         | E <sub>γ</sub> : from 1974Be18.<br>I <sub>γ</sub> : weighted average of 8 5 (1974Be18), 10 7 (1972Ca22), and 15 5 (1974Gr19).                                                                                                                                                                      |

Continued on next page (footnotes at end of table)

$^{32}\text{S}(^3\text{He},n\gamma)$  **1974Gr19,1974Be18,1972Ca22 (continued)** $\gamma(^{34}\text{Ar})$  (continued)

| $E_i(\text{level})$ | $J_i^\pi$ | $E_\gamma^\dagger$ | $I_\gamma^\ddagger$ | $E_f$  | $J_f^\pi$ | Comments                                                                      |
|---------------------|-----------|--------------------|---------------------|--------|-----------|-------------------------------------------------------------------------------|
| 3873                | $0^+$     | 585                | <30                 | 3287.5 | $2^+$     | $I_\gamma$ : <30 (1974Be18).                                                  |
|                     |           | 1782.3             | >70                 | 2091.0 | $2^+$     | $E_\gamma, I_\gamma$ : from 1974Be18; other: $I_\gamma=100$ (1974Gr19).       |
| 4127.4              | $(2^+)$   | 839.9              | 90 5                | 3287.5 | $2^+$     | $E_\gamma$ : weighted average of 840.1 9 (1974Be18) and 839.1 18 (1972Ca22).  |
|                     |           |                    |                     |        |           | $I_\gamma$ : others: >50 (1974Be18), 100 (1972Ca22).                          |
|                     |           |                    |                     |        |           | $\delta$ : +0.52 15 from $n\gamma(\theta)$ if $J_i=2$ .                       |
|                     |           | 2037               | 10 5                | 2091.0 | $2^+$     | $I_\gamma$ : other: <50 (1974Be18).                                           |
|                     |           | 4128               | <10                 | 0.0    | $0^+$     |                                                                               |
| 4513.5              | 3         | 1226.0             | 90 10               | 3287.5 | $2^+$     | $E_\gamma$ : weighted average of 1225.5 6 (1974Be18) and 1226.7 7 (1972Ca22). |
|                     |           |                    |                     |        |           | $I_\gamma$ : others: >90 (1974Be18), 100 (1972Ca22).                          |
|                     |           |                    |                     |        |           | $A_2=-0.38$ 24 (1974Be18).                                                    |
|                     |           | 2422               | 10 3                | 2091.0 | $2^+$     | $I_\gamma$ : others: <10 (1974Be18), <20 (1972Ca22).                          |
| 4631                |           | 2540               | >50                 | 2091.0 | $2^+$     |                                                                               |
| 4865                |           | 2774               | >50                 | 2091.0 | $2^+$     |                                                                               |
| 4967                |           | 841                | <20                 | 4127.4 | $(2^+)$   |                                                                               |
|                     |           | 2876               | >50                 | 2091.0 | $2^+$     |                                                                               |
| 5255                |           | 3164               | >50                 | 2091.0 | $2^+$     |                                                                               |
| 5542                |           | 911                | 35 10               | 4631   |           |                                                                               |
|                     |           | 1029               | 65 10               | 4513.5 | 3         |                                                                               |

 $^\dagger$  Values without uncertainties are from level-energy differences. $^\ddagger$  From 1974Gr19, unless otherwise noted.

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Level Scheme

Intensities: % photon branching from each level

